

# Joshua Oh Yu Qiao

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Available: 12-week placement, 25 January – 23 April 2027

## SUMMARY

Third-year Electronic and Electrical Engineering student (CGPA 3.89) who takes embedded products end to end — C/C++ firmware on ESP32/ESP8266, sensor interfacing over I<sup>2</sup>C/I<sup>2</sup>S/UART, custom circuits and 3D-printed enclosures. Three university projects carried from schematic to working hardware, including a solar-powered field controller characterised at 0.89 W idle / 12.8 W actuating.

## EDUCATION

**Bachelor of Electronic and Electrical Engineering with Honours** Apr 2024 – Apr 2028  
*Sunway University, Subang Jaya*

- Current CGPA: 3.89 / 4.00 — Dean's List: Sep 2024, Apr 2025, Sep 2025 semesters
- Jeffrey Cheah Entrance Scholarship recipient

**Cambridge A-Levels — 3A (Mathematics, Physics, Chemistry)** Jul 2022 – Nov 2023  
*INTI International College Subang — Top Student, A-Level (2023) & AS-Level (2023)*

**Sijil Pelajaran Malaysia (SPM), Science Stream — 8A, 1B** Jan 2017 – Mar 2022  
*SMK Bandar Sri Damansara 1, Kuala Lumpur*

## ENGINEERING PROJECTS

### AI Waste-Sorting Prototype

*Sustainable Engineering Design, group project*

- Trained and deployed a custom YOLOv11s model on a self-collected 110-image dataset to classify metal, paper and plastic in real time, achieving 0.98 mAP@0.5 and 0.96 F1 at 0.63 confidence.
- Integrated the vision pipeline with an ESP32 over HTTP to drive a conveyor and dual servo diverters, with three HX711 load cells for per-bin weight logging, a Blynk dashboard and automated bin-full alerts.
- Modelled the mechanical assembly in SOLIDWORKS and 3D-printed it in PLA (242 g, 23 × 18 × 19 cm); designed the circuit in EasyEDA and assembled it on perfboard.

### PIPER — Privacy-Preserving Dementia Assistive Robot

*Integrated Design Project, 4-person team — System Architect & Firmware Lead · prototype build underway*

- Architected a two-tier system pairing an ESP32 with a Python/Flask backend running local Whisper STT, Gemini LLM and Piper TTS over an SQLite event-memory database, enabling the robot to recall the user's daily events (medication, movement, appliance state).
- Ran a comparative literature review of fall-detection sensing methods and selected a 24 GHz mmWave radar (LD2450) over camera-based alternatives, eliminating video capture in the home while preserving day/night reliability.
- Built Telegram caregiver alerting with a 10-second cancellation window and MQTT smart-home control across two ESP32 nodes, designed to a <3 s voice-response and <10 s alert budget.

### Solar-Powered IoT Irrigation Controller

*Project Management & Engineering Design — Project Manager and Software Engineer, 5-person team*

- Characterised the Li-ion power system at 0.89 W idle / 12.8 W actuating (~1.7 days runtime per charge); diagnosed and suppressed solenoid back-EMF with a flyback diode after it caused relay chatter.
- Built the firmware on an ESP8266 in C++ (VS Code / PlatformIO / Git), reading a DS18B20 probe to drive relay-switched solenoid valve actuation above a 38 °C threshold — preventing thermal shock to crops — with a Blynk dashboard for remote auto/manual control and NTP-synced logging.
- Led the team to delivery on a 13-week CPM/Gantt schedule and 9% under budget, running critical-path analysis and budget variance reporting.

## TECHNICAL SKILLS

- **Programming:** C/C++, Python (NumPy, Pandas, OpenCV, PyTorch/Ultralytics), MATLAB & Simulink, Git/GitHub, VS Code, PlatformIO, Arduino IDE
- **Embedded & Hardware:** ESP32 / ESP8266 (NodeMCU), sensor interfacing (mmWave radar, DS18B20, HX711 load cells, I<sup>2</sup>S MEMS microphones), servo & relay actuation, I<sup>2</sup>C / I<sup>2</sup>S / UART / OneWire, voltage regulation and power budgeting, soldering and perfboard prototyping, oscilloscopes and multimeters
- **IoT, Cloud & AI:** MQTT, Flask REST APIs, SQLite, HTTP/REST device communication, LLM and speech-pipeline integration (Gemini, Whisper, Piper TTS), Blynk, ThingSpeak, SMTP/Telegram alerting, real-time dashboards and data logging
- **Design & Simulation:** SOLIDWORKS, OnShape, EasyEDA, NI Multisim, FDM 3D printing (PLA/ABS)
- **Engineering Practice:** Project management (CPM, Gantt, budget variance analysis), decision-matrix design selection, laboratory instrumentation, technical report writing
- **Languages:** Bahasa Malaysia, English, Mandarin, Cantonese (fluent); German (basic)

## WORK EXPERIENCE

**Assistant Head, Events & Social Media — Sales Support Team (Part-time)** Feb 2023 – Apr 2024  
*INTI International College Subang, Subang Jaya*

- Led a team of 8 to 12 student helpers in various events (Mid Valley Higher Education Fair, INTI Open Day, WOW Event).
- Involved in the recruitment process for events with the HR Team.
- Conducted meetings and training sessions for student helpers ahead of events.
- Conducted Data Entry and rescue calls to help with leads collection.